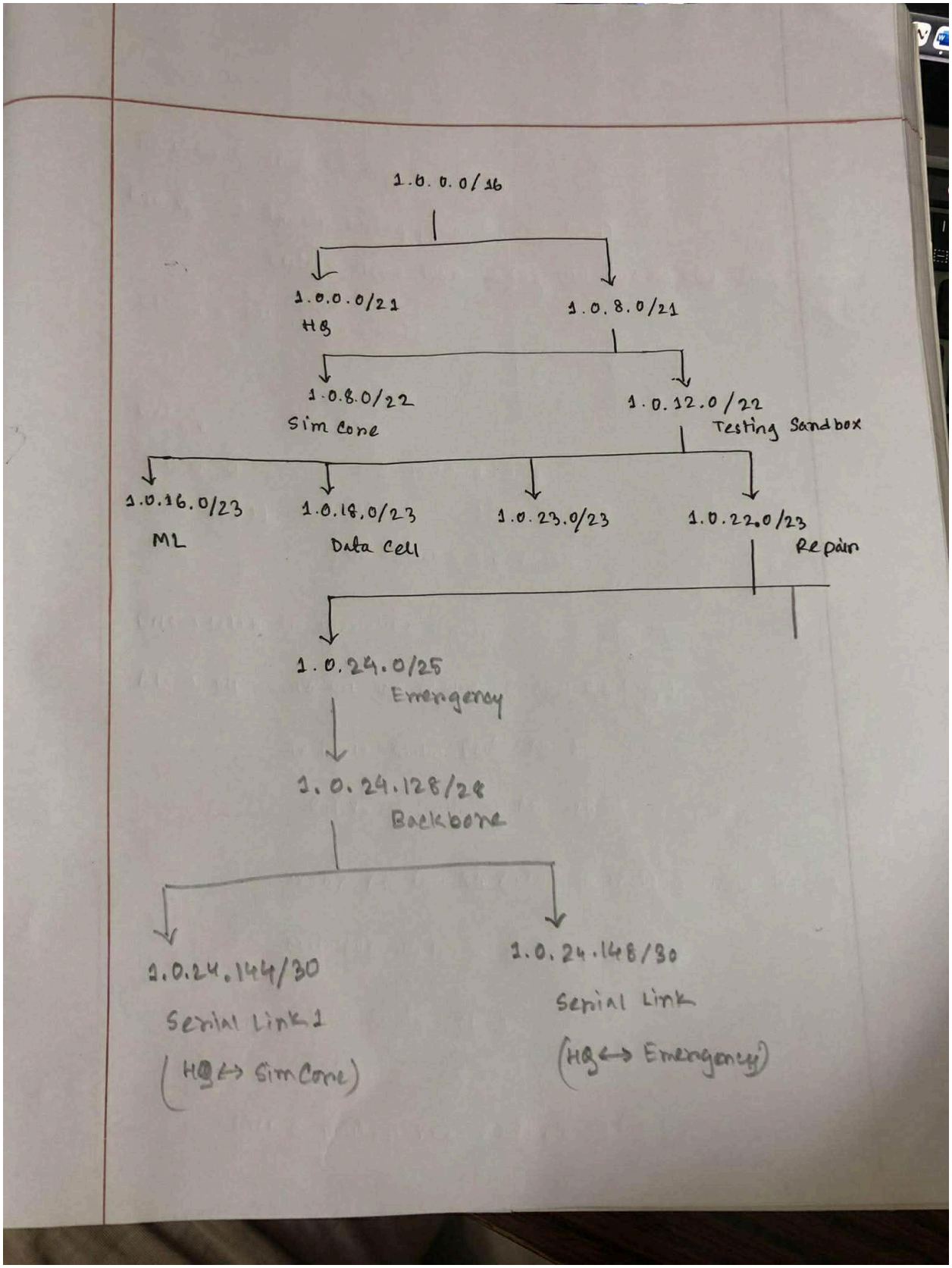
IP Addressing Table (Routers)

Router	Interface	IP Address	Mask	Purpose
R_HQ	Fa0/0	1.0.0.1	255.255.248.0	HQ Control Wing LAN (gateway)
R_HQ	S0/0/0	1.0.24.145	255.255.255.252	To R_SimCore (DCE set clock rate)
R_HQ	S0/0/1	1.0.24.149	255.255.255.252	To R_Emergency (DCE set clock rate)
R_SimCore	Fa0/0	1.0.8.1	255.255.252.0	Simulation Core LAN (gateway)
R_SimCore	Fa0/1	1.0.24.129	255.255.255.240	Backbone switch
R_SimCore	S0/0/0	1.0.24.146	255.255.255.252	To R_HQ (DTE no clock rate)
R_SimSandbox	Fa0/0	1.0.12.1	255.255.252.0	Testing Sandbox LAN (gateway)
R_SimSandbox	Fa0/1	1.0.24.130	255.255.255.240	Backbone switch
R_ML	Fa0/0	1.0.16.1	255.255.254.0	ML Engine LAN (gateway)
R_ML	Fa0/1	1.0.24.131	255.255.255.240	Backbone switch
R_DP	Fa0/0	1.0.18.1	255.255.254.0	Data Processing Cell LAN (gateway)
R_DP	Fa0/1	1.0.24.132	255.255.255.240	Backbone switch
R_Diag	Fa0/0	1.0.20.1	255.255.254.0	Diagnostics Bay LAN (gateway)
R_Diag	Fa0/1	1.0.24.133	255.255.255.240	Backbone switch
R_Repair	Fa0/0	1.0.22.1	255.255.254.0	Repair Unit LAN (gateway)
R_Repair	Fa0/1	1.0.24.134	255.255.255.240	Backbone switch
R_Emergency	Fa0/0	1.0.24.1	255.255.255.128	Backup Core LAN (gateway)
R_Emergency	S0/0/0	1.0.24.150	255.255.255.252	To R_HQ (DTE no clock rate)

VLSM Tree


1.0.0.0/16
1.0.0.0/21
HQ
1.0.8.0/21
1.0.8.0/22
Sim Core
1.0.12.0/22
Testing Sandbox
1.0.16.0/23
ML
1.0.18.0/23
Data Cell
1.0.23.0/23
1.0.22.0/23
Repair
1.0.24.0/25
Emergency
1.0.24.128/28
Backbone
1.0.24.144/30
Serial Link 1
(HQ ↔ Sim Core)
1.0.24.148/30
Serial Link
(HQ ↔ Emergency)

Commands:

Command---R_HQ

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_HQ
R_HQ(config)#no ip domain-lookup
R_HQ(config)#interface FastEthernet0/0
R_HQ(config-if)#ip address 1.0.0.1 255.255.248.0
R_HQ(config-if)#no shutdown

R_HQ(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R_HQ(config-if)#exit
R_HQ(config)#interface Serial0/0/0
R_HQ(config-if)#ip address 1.0.24.145 255.255.255.252
R_HQ(config-if)#clock rate 64000
R_HQ(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to down
R_HQ(config-if)#exit

R_HQ(config)#interface Serial0/0/1
R_HQ(config-if)#ip address 1.0.24.149 255.255.255.252
R_HQ(config-if)#clock rate 64000
R_HQ(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0/1, changed state to down
R_HQ(config-if)#exit
R_HQ(config)#ip route 1.0.24.0 255.255.255.128 1.0.24.150
R_HQ(config)#ip route 1.0.8.0 255.255.252.0 1.0.24.146
R_HQ(config)#ip route 1.0.12.0 255.255.252.0 1.0.24.146
R_HQ(config)#ip route 1.0.16.0 255.255.254.0 1.0.24.146
R_HQ(config)#ip route 1.0.18.0 255.255.254.0 1.0.24.146
```

```
R_HQ(config)#ip route 1.0.20.0 255.255.254.0 1.0.24.146
R_HQ(config)#ip route 1.0.22.0 255.255.254.0 1.0.24.146
R_HQ(config)#ip route 1.0.24.128 255.255.255.240 1.0.24.146
R_HQ(config)#end
R_HQ#
%SYS-5-CONFIG_I: Configured from console by console

R_HQ#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Command --- R_SimCore

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_SimCore
R_SimCore(config)#no ip domain-lookup
R_SimCore(config)#interface FastEthernet0/0
R_SimCore(config-if)#ip address 1.0.8.1 255.255.252.0
R_SimCore(config-if)#no shutdown

R_SimCore(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R_SimCore(config)#interface FastEthernet0/1
R_SimCore(config-if)#ip address 1.0.24.129 255.255.255.240
R_SimCore(config-if)#no shutdown

R_SimCore(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R_SimCore(config)#interface Serial0/0/0
R_SimCore(config-if)#ip address 1.0.24.146 255.255.255.252
R_SimCore(config-if)#no shutdown

R_SimCore(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
```

```
exit
R_SimCore(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R_SimCore(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.145
R_SimCore(config)#router rip
R_SimCore(config-router)#version 2
R_SimCore(config-router)#no auto-summary
R_SimCore(config-router)#network 1.0.0.0
R_SimCore(config-router)#passive-interface FastEthernet0/0
R_SimCore(config-router)#passive-interface Serial0/0/0
R_SimCore(config-router)#exit
R_SimCore(config)#end
R_SimCore#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
R_SimCore#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Command---R_SimSandbox

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_SimSandbox
R_SimSandbox(config)#no ip domain-lookup
R_SimSandbox(config)#interface FastEthernet0/0
R_SimSandbox(config-if)#ip address 1.0.12.1 255.255.252.0
R_SimSandbox(config-if)#ip helper-address 1.0.8.2
R_SimSandbox(config-if)#no shutdown

R_SimSandbox(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R_SimSandbox(config)#interface FastEthernet0/1
R_SimSandbox(config-if)#ip address 1.0.24.130 255.255.255.240
R_SimSandbox(config-if)#no shutdown
```

```
R_SimSandbox(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R_SimSandbox(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_SimSandbox(config)#router rip
R_SimSandbox(config-router)#version 2
R_SimSandbox(config-router)#no auto-summary
R_SimSandbox(config-router)#network 1.0.0.0
R_SimSandbox(config-router)#passive-interface FastEthernet0/0
R_SimSandbox(config-router)#exit
R_SimSandbox(config)#end
R_SimSandbox#
%SYS-5-CONFIG_I: Configured from console by console

R_SimSandbox#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R_SimSandbox#
```

Command---R_ML

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_ML
R_ML(config)#no ip domain-lookup
R_ML(config)#interface FastEthernet0/0
R_ML(config-if)#ip address 1.0.16.1 255.255.254.0
Bad mask 0xFFFFF400 for address 1.0.16.1
R_ML(config-if)#no shutdown

R_ML(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R_ML(config)#interface FastEthernet0/1
R_ML(config-if)#ip address 1.0.24.131 255.255.255.240
R_ML(config-if)#no shutdown

R_ML(config-if)#
```

```
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R_ML(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_ML(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.130 84
R_ML(config)#ip dhcp excluded-address 1.0.16.1 1.0.16.10
R_ML(config)#ip dhcp pool ML_Engine_Pool
R_ML(dhcp-config)#network 1.0.16.0 255.255.254.0
R_ML(dhcp-config)#default-router 1.0.16.1
R_ML(dhcp-config)#dns-server 1.0.0.2
R_ML(dhcp-config)#exit
R_ML(config)#router rip
R_ML(config-router)#version 2
R_ML(config-router)#no auto-summary
R_ML(config-router)#network 1.0.0.0
R_ML(config-router)#passive-interface FastEthernet0/0
R_ML(config-router)#exit
R_ML(config)#end
R_ML#
%SYS-5-CONFIG_I: Configured from console by console

R_ML#write memory
Building configuration...
[OK]
R_ML#
R_ML#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```

Command---R_DP

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_DP
R_DP(config)#no ip domain-lookup
R_DP(config)#interface FastEthernet0/0
R_DP(config-if)#ip address 1.0.18.1 255.255.254.0
R_DP(config-if)#no shutdown

R_DP(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
```

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R_DP(config)#interface FastEthernet0/1
R_DP(config-if)#ip address 1.0.24.132 255.255.255.240
R_DP(config-if)#no shutdown

R_DP(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R_DP(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_DP(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_DP(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.130 84
R_DP(config)#ip dhcp excluded-address 1.0.18.1 1.0.18.10
R_DP(config)#ip dhcp pool DP_Cell_Pool
R_DP(dhcp-config)#network 1.0.18.0 255.255.254.0
R_DP(dhcp-config)#default-router 1.0.18.1
R_DP(dhcp-config)#dns-server 1.0.0.2
R_DP(dhcp-config)#exit
R_DP(config)#router rip
R_DP(config-router)#version 2
R_DP(config-router)#no auto-summary
R_DP(config-router)#network 1.0.0.0
R_DP(config-router)#passive-interface FastEthernet0/0
R_DP(config-router)#exit
R_DP(config)#end
R_DP#
%SYS-5-CONFIG_I: Configured from console by console

R_DP#write memory
Building configuration...
[OK]
R_DP#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R_DP#
```

Command---R_Diag

```
Router>en
```

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_Diag
R_Diag(config)#no ip domain-lookup
R_Diag(config)#interface FastEthernet0/0
R_Diag(config-if)#ip address 1.0.20.1 255.255.254.0
R_Diag(config-if)#no shutdown

R_Diag(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R_Diag(config)#interface FastEthernet0/1
R_Diag(config-if)#ip address 1.0.24.133 255.255.255.240
R_Diag(config-if)#no shutdown

R_Diag(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R_Diag(config-if)#exit
R_Diag(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_Diag(config)#ip dhcp excluded-address 1.0.20.1 1.0.20.10
R_Diag(config)#ip dhcp pool Diag_Bay_Pool
R_Diag(dhcp-config)#network 1.0.20.0 255.255.254.0
R_Diag(dhcp-config)#default-router 1.0.20.1
R_Diag(dhcp-config)#dns-server 1.0.0.2
R_Diag(dhcp-config)#exit
R_Diag(config)#router rip
R_Diag(config-router)#version 2
R_Diag(config-router)#no auto-summary
R_Diag(config-router)#network 1.0.0.0
R_Diag(config-router)#passive-interface FastEthernet0/0
R_Diag(config-router)#exit
R_Diag(config)#
R_Diag(config)#end
R_Diag#
%SYS-5-CONFIG_I: Configured from console by console

R_Diag#write memory
Building configuration...
[OK]
```

```
R_Diag#
R_Diag#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R_Diag#
R_Diag#
```

Command---R_Repair

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R_Repair
R_Repair(config)#no ip domain-lookup
R_Repair(config)#interface FastEthernet0/0
R_Repair(config-if)#ip address 1.0.22.1 255.255.254.0
R_Repair(config-if)#no shutdown

R_Repair(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R_Repair(config-if)#exit
R_Repair(config)#interface FastEthernet0/1
R_Repair(config-if)#ip address 1.0.24.134 255.255.255.240
R_Repair(config-if)#no shutdown

R_Repair(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

R_Repair(config-if)#exit
R_Repair(config)#ip route 1.0.0.0 255.255.248.0 1.0.24.129
R_Repair(config)#ip dhcp excluded-address 1.0.22.1 1.0.22.10
R_Repair(config)#ip dhcp pool Repair_Unit_Pool
R_Repair(dhcp-config)#network 1.0.22.0 255.255.254.0
R_Repair(dhcp-config)#default-router 1.0.22.1
R_Repair(dhcp-config)#dns-server 1.0.0.2
R_Repair(dhcp-config)#exit
R_Repair(config)#
```

```
R_Repair(config)#router rip
R_Repair(config-router)#version 2
R_Repair(config-router)#no auto-summary
R_Repair(config-router)#network 1.0.0.0
R_Repair(config-router)#passive-interface FastEthernet0/0
R_Repair(config-router)#exit
R_Repair(config)#
R_Repair(config)#end
R_Repair#
%SYS-5-CONFIG_I: Configured from console by console

R_Repair#write memory
Building configuration...
[OK]
R_Repair#
R_Repair#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R_Repair#
```

Command---R_Emergency

```
Router>
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#hostname R_Emergency
R_Emergency(config)#no ip domain-lookup
R_Emergency(config)#interface FastEthernet0/0
R_Emergency(config-if)#ip address 1.0.24.1 255.255.255.128
R_Emergency(config-if)#no shutdown

R_Emergency(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R_Emergency(config-if)#exit
R_Emergency(config)#interface Serial0/0/0
R_Emergency(config-if)#ip address 1.0.24.150 255.255.255.252
R_Emergency(config-if)#no shutdown
```

```
R_Emergency(config-if)#
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
exit
R_Emergency(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

R_Emergency(config)#ip route 1.0.0.0 255.255.248.0 Serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
R_Emergency(config)#end
R_Emergency#
%SYS-5-CONFIG_I: Configured from console by console

R_Emergency#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R_Emergency#
R_Emergency#
```

List of any assumptions we made:

We used (.com) instead of (.federal)